

Summer Project Proposal Report On
Artisa: An E-Commerce Platform for Artists to Showcase,
Sell, and Commission Artwork

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ABSTRACT

The rapid growth of digital technologies and e-commerce has created new opportunities for artists to showcase, sell, and promote their artwork beyond traditional galleries and exhibitions. However, many artists still depend on multiple platforms for portfolio management, marketing, artwork sales, commission services, communication, and payment processing, resulting in a fragmented and inefficient workflow. Additionally, the increasing use of generative Artificial Intelligence (AI) has raised concerns regarding originality, copyright, and the recognition of authentic human-created artwork, increasing the need for trusted platforms that support verified artists.

This project proposes **Artisa**, a web-based multi-vendor e-commerce platform designed to provide a centralized marketplace for artists and art enthusiasts. The platform aims to integrate artwork buying and selling, artist portfolio management, commission requests, customer–artist interaction, and secure payment processing within a single system. Users will be able to register as customers and apply to become verified artists, while administrators will manage artist verification, platform content, and transactions.

The system will be developed using **React.js and Tailwind CSS** for the frontend, **PHP with Laravel Framework** for backend development, and **MySQL** for database management. Additional tools such as Figma, GitHub, Postman, and Jira will support system design, development, testing, and project management. The expected outcome of this project is a secure and user-friendly platform that simplifies the online workflow of artists while providing customers with a reliable environment to discover, purchase, and commission original artwork, supporting the growth of Nepal's digital creative economy.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BIM	Bachelor of Information Management
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTP	Hypertext Transfer Protocol
IDE	Integrated Development Environment
MVC	Model-View-Controller
MySQL	My Structured Query Language
PHP	Hypertext Preprocessor
SQL	Structured Query Language
TU	Tribhuvan University
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

CHAPTER I: INTRODUCTION

1.1 Introduction

Art has always been an important medium of creativity, culture, and self-expression not just a decoration especially in a country like Nepal playing a significant role in preserving cultural heritage and supporting the creative economy. With the rapid growth of the internet and digital technologies, artists are no longer limited to galleries and exhibitions to showcase or sell their artwork. Today e-commerce and online platforms allow artists to reach wider audiences, sell artwork, and accept commissions from customers from anywhere in the world.

In Nepal, the increasing use of the internet, social media, and digital payment services has encouraged many artists to promote and sell their work online creating new opportunities to reach both local and international customers. Despite these developments, there is still an opportunity for platforms that better support the needs of artists through integrated digital services.

To address these challenges, this project proposes **Artisa**, a web-based multi-vendor e-commerce platform designed specifically for artists and art enthusiasts. The platform integrates artwork sales, artist portfolios, commission management, customer communication, and secure payment processing within a single system. The platform also includes a Verified Artist system to improve trust between artists and customers while promoting original creative work.

Overall, Artisa aims to provide a centralized e-commerce platform that simplifies the online workflow of artists while offering customers a secure and convenient environment to discover, purchase, and commission original artwork. By integrating multiple services into a single platform, the system seeks to support Nepal's growing creative economy and create new opportunities for artists to reach both local and international markets.

1.2 Problem Statement

Although e-commerce and digital technologies have created new opportunities for artists, Nepal still lacks a dedicated platform that integrates artist portfolios, artwork sales, commission services, and secure online transactions. Existing platforms such as **Daraz Nepal**, **Hamrobazar**, and **Facebook Marketplace** support general buying and selling but are not designed to meet the specific needs of artists.

As a result, artists often use multiple platforms for different purposes, such as **Facebook** and **Instagram** for promotion, **Behance** for portfolios, **Fiverr** for commissions, and messaging applications for customer communication. Managing multiple platforms is time-consuming, inefficient, and can make it difficult for artists to organize their work and maintain consistent interactions with customers.

Furthermore, the increasing use of generative Artificial Intelligence (AI) has raised concerns about originality and the visibility of human-created artwork. While AI has introduced new creative possibilities, customers may find it more difficult to distinguish authentic artists from anonymous online sellers, increasing the importance of trusted platforms that promote verified creators and transparent transactions.

Therefore, there is a need for a specialized e-commerce platform that brings together artwork sales, portfolio management, commission services, customer communication, and secure payment processing within a single system. Such a platform can simplify the workflow for artists, improve the purchasing experience for customers, and contribute to the growth of Nepal's digital creative economy.

1.3 Objectives

1.3.1 General Objective

To design and develop Artisa, a secure web-based multi-vendor e-commerce platform that enables Nepali artists to showcase portfolios, sell physical and digital artwork, receive commission requests, and connect with customers through an integrated online marketplace.

1.3.2 Specific Objectives

The project aims to:

- Develop a centralized marketplace for buying, selling, and commissioning physical and digital artwork.
- Enable artists to create professional portfolios and apply for verified artist status.
- Provide customers with secure artwork purchasing, commission requests, and integrated communication with artists.
- Integrate secure online payment gateways for transactions.
- Implement search, filtering, and categorization features to improve artwork and artist discovery.
- Develop an administrative system to manage users, verify artists, moderate content, and monitor transactions.
- Design a responsive, user-friendly web application that supports the growth of Nepal's digital creative industry.

1.4 Development Methodology

The development of **Artisa** will follow the **Agile Software Development Methodology**, specifically the Scrum approach. Agile is an iterative software development methodology that divides a project into smaller development cycles called sprints. Each sprint focuses on implementing and testing a specific set of features, allowing continuous improvement throughout the development process.

Agile has been selected because the requirements of the proposed system may evolve during development based on feedback from the project supervisor and system testing. Instead of completing the entire system at once, the project will be developed in manageable stages, making it easier to identify issues, improve functionality, and enhance the user experience.

The development process of Artisa will consist of the following phases:

- **Requirement Analysis:** Identify functional and non-functional requirements through research and system planning.
- **System Design:** Prepare system architecture, database design, user interface prototypes, and UML diagrams.
- **Development:** Implement the frontend, backend, database, and core functionalities such as user management, artwork listings, commission requests, and payment integration.
- **Testing:** Perform functional testing and resolve identified issues to ensure the system operates correctly.
- **Deployment and Evaluation:** Deploy the completed system for demonstration and evaluate whether it meets the project objectives.

The Agile methodology provides flexibility throughout the development process and supports continuous improvement, making it suitable for developing a user-centered e-commerce platform such as Artisa.

CHAPTER II: METHODOLOGY

2.1 Requirement Identification and Feasibility Study

2.1.1 Requirement Identification

The requirements for the proposed system were identified through the study of existing e-commerce platforms, artist portfolio websites, research articles, and discussions with potential users. Platforms such as **Etsy**, **Behance**, **Fiverr**, **Daraz Nepal**, and **Hamrobazar** were reviewed to understand their features, strengths, and limitations. The findings were used to identify the functional and non-functional requirements of Artisa and to determine how the proposed system could better support artists and customers.

Based on the requirement analysis, the proposed system will include the following:

a. Functional Requirements

- User registration and secure login.
- Artist verification by the administrator.
- Artist portfolio management.
- Upload and management of physical and digital artwork.
- Artwork search, filtering, and categorization.
- Shopping cart and order management.
- Commission request and approval system.
- Integrated communication between artists and customers.
- Secure online payment gateway integration.
- User reviews and ratings.
- Administrative dashboard for managing users, artists, artwork, and transactions.

b. Non-Functional Requirements

- Responsive user interface for desktop and mobile devices.
- Secure authentication and data protection.
- Fast system performance and efficient database management.
- User-friendly interface with intuitive navigation.
- Reliable system availability and maintainability.
- Scalable architecture to support future feature expansion.

2.1.2 Feasibility Study

A feasibility study was conducted to evaluate whether the proposed system could be successfully developed within the available resources and project duration.

a. Technical Feasibility

The project is technically feasible because it will be developed using widely adopted web technologies, including **React.js** with **Tailwind CSS** for the frontend, **PHP with Laravel Framework** for the backend, and **MySQL** for database management.

Development tools such as **Visual Studio Code**, **GitHub**, and **Postman** provide the necessary support for implementation, testing, and version control. The required software is freely available, and the developer has access to a personal computer capable of supporting the development environment.

b. Economic Feasibility

The proposed system requires minimal development cost because most of the software, frameworks, and development tools are open source or available free of charge for educational purposes. The primary investment is the developer's time, with optional expenses such as domain registration or web hosting if the system is deployed online.

c. Operational Feasibility

The system is operationally feasible because it is designed to simplify the workflow of artists while providing customers with a secure and convenient marketplace. The interface will be designed to be easy to use, requiring minimal technical knowledge from users. Administrators will be able to manage artist verification, platform content, and transactions through a centralized dashboard.

d. Schedule Feasibility

The project is expected to be completed within one academic semester. The development process will follow the Agile methodology, allowing the system to be implemented incrementally through planning, design, development, testing, and evaluation. This approach makes the project achievable within the available timeframe while allowing improvements based on supervisor feedback.

2.2 Related Work / Literature Review

The growth of e-commerce has significantly changed how artists promote, sell, and distribute their work online. Various digital platforms have been developed to support online commerce, portfolio presentation, and freelance services. However, most existing platforms are designed to fulfill specific purposes rather than providing a complete solution for artists.

Many existing platforms provide services that support different aspects of an artist's workflow. **Behance** allow artists to create professional portfolios and showcase their work, while Etsy provides an online marketplace for selling handmade and digital products. **Fiverr** focuses on commission-based freelance services, enabling artists to offer customized artwork to clients. In Nepal, platforms such as **Daraz Nepal** and **Hamrobazar** support online buying and selling but are designed for general commerce rather than the specific needs of artists.

Recent studies have also highlighted the growing influence of generative Artificial Intelligence (AI) on the creative industry. Although AI has introduced new creative possibilities, it has also raised concerns regarding originality, copyright, and the visibility of authentic human-created artwork. These challenges increase the importance of platforms that promote verified artists and transparent transactions.

Research on digital marketplaces emphasizes that successful e-commerce platforms should provide secure transactions, efficient marketplace management, and trustworthy interactions between buyers and sellers. Studies also indicate that online art platforms continue to rely on multiple intermediaries and fragmented workflows, highlighting opportunities for systems that better integrate artist portfolios, artwork sales, and customer engagement.

Based on the reviewed literature and existing platforms, it can be observed that no single platform fully integrates portfolio management, artwork sales, commission requests, artist verification, customer communication, and secure online transactions within one system. This gap provides the motivation for the proposed Artisa platform, which aims to offer a unified marketplace tailored to the needs of artists and customers.

Table 1.1 Comparison of Existing Platforms and the Proposed System

The table below shows that existing platforms provide useful features for artists and online commerce but generally focus on specific areas. Platforms such as Behance mainly support portfolio presentation, while Etsy and Fiverr focus on selling and commission-based services. However, these platforms do not provide a complete integrated solution combining artwork sales, portfolio management, commission handling, artist verification, and customer communication. This identified gap supports the need for the proposed Artisa platform.

Platform	Core Focus	Strengths	Limitations for Artists
Daraz Nepal	General e-commerce marketplace	Large user base, logistics network	No portfolio, commission, or artist-verification features
Hamrobazar/Facebook Marketplace	Classified buying & selling	Free listing, wide local reach	No secure payment, no portfolio tools, high scam risk
Behance	Portfolio showcase	Strong visual presentation, community	No integrated sales or payment system
Fiverr/Upwork	Freelance commission marketplace	Structured commission workflow, escrow payment	Generic freelancing focus, high service fees, not Nepal-specific
Etsy	Global handmade & art marketplace	Established international buyer base	Not localized for Nepal, no local payment gateway support

2.3 Analysis and Design Tools

The analysis and design phase focuses on understanding system requirements, planning the system structure, and designing the user interface before implementation. Various modeling and design tools will be used to ensure that the proposed Artisa platform is properly structured, user-friendly, and maintainable.

Requirement Analysis Tools

UML (Unified Modeling Language) will be used to represent the functional structure of the system. UML diagrams help visualize system components, user interactions, and workflows before development. The following diagrams will be prepared:

- **Use Case Diagram:** To represent interactions between system users (customers, artists, and administrators) and system functionalities.
- **Activity Diagram:** To describe workflows such as artwork purchasing, commission requests, and artist verification processes.
- **Sequence Diagram:** To illustrate communication flow between users and system components during important operations.
- **Class Diagram:** To represent the structure of system entities and their relationships.

Data Flow Diagram (DFD) will be used to represent the movement of data within the system. It will illustrate how information flows between users, system processes, and databases. DFDs will help analyze major processes such as user management, artwork uploads, order processing, commission handling, and payment transactions.

Database Design Tool

An **Entity Relationship Diagram (ERD)** will be created to design the database structure of Artisa. The ERD will define relationships between major entities such as

users, artists, artworks, orders, commissions, payments, reviews, and transaction records. This will help ensure efficient data organization and reduce database design errors.

User Interface Design Tool

Figma will be used for creating wireframes and user interface prototypes. The designs will help plan the layout, navigation, and user experience of the platform before development. The prototypes will include major interfaces such as:

- User registration and login pages.
- Artwork detail pages.
- Artist portfolio pages.
- Artwork marketplace pages.
- Wishlist and order history pages.
- Shopping cart and checkout pages.
- Commission request pages.
- Administrator and Artist dashboard.

System Design Approach

A three-tier architecture approach will be followed for designing the system:

- **Presentation Layer:** Responsible for user interaction through the frontend interface.
- **Application Layer:** Handles business logic and communication between different system components.
- **Data Layer:** Manages storage and retrieval of user information, artwork details, transactions, and system records.

These analysis and design tools will help create a structured development plan and ensure that the Artisa platform meets user requirements while maintaining usability, security, and scalability.

2.4 Implementation Tools

The implementation of Artisa will involve frontend development, backend development, database management, file storage, authentication, and system testing. The following technologies and tools will be used for developing the proposed platform.

Frontend Development

React.js will be used for developing the frontend interface of Artisa. React provides a component-based structure that allows the creation of reusable and maintainable user interface components. **Tailwind CSS** will be used to create responsive and modern layouts for customers, artists, and administrators.

Backend Development

PHP with Laravel Framework will be used for backend development. Laravel provides an organized MVC architecture, built-in security features, database management support, authentication handling, and efficient API development. It will manage system operations including user management, artwork management, commission requests, order processing, and administrative functions.

Database Management

MySQL will be used as the database management system for storing and managing structured information including user accounts, artist profiles, artworks, orders, commissions, payments, reviews, and transaction records.

File Storage

Cloudinary will be used for managing artwork images and digital files uploaded by artists. It provides optimized media storage and allows efficient handling of large image files without directly storing them in the database.

Authentication and Security

Laravel Sanctum authentication system will be used along with role-based access control to manage different types of users, including customers, verified artists, and administrators. Password security and user authorization will be implemented to protect user information and system resources.

Development and Testing Tools

The following tools will be used during development:

- **Visual Studio Code:** Source code development and project management.
- **Git and GitHub:** Version control and backup of project files.
- **Postman:** Testing and validating backend APIs.
- **Figma:** UI/UX design and prototype creation.
- **Jira:** Used to manage development tasks, track project progress, and organize work iterations according to the Agile methodology.

Payment Integration

A secure online payment gateway will be integrated to support artwork purchases and approved commission payments. The selected payment service will be based on availability and suitability for the developed system.

The selected technologies provide a practical foundation for developing Artisa within the available semester timeframe while ensuring maintainability, security, and scalability. (*Khalti Payment Gateway*, n.d.)

CHAPTER III: EXPECTED OUTCOME

3.1 Expected Outcomes

The proposed Artisa platform is expected to provide a centralized web-based marketplace that connects artists and customers through a secure and organized digital environment. The system will simplify the process of showcasing, purchasing, and commissioning artwork by integrating multiple functionalities into a single platform.

The expected outcomes of this project include:

- A functional multi-vendor e-commerce platform where artists can sell physical and digital artwork.
- A portfolio management system allowing artists to showcase their creative work professionally.
- A verified artist system that improves customer trust and promotes authentic artwork.
- A commission management system that allows customers to request customized artwork from artists.
- A product browsing system with categories, search, and filtering features for easier artwork discovery.
- A shopping cart and order management system for purchasing artworks.
- A secure payment integration system for online transactions.
- A communication system that allows interaction between artists and customers during commission processes.
- A centralized database for managing users, artworks, orders, payments, reviews, and transaction history.
- An administrative dashboard for managing users, artist verification, artworks, categories, and platform activities.

Overall, Artisa aims to provide a user-friendly platform that reduces the dependency on multiple separate platforms and supports the growth of Nepal's digital creative marketplace by providing artists with better opportunities to reach customers locally and internationally.

3.2 Project Schedule

Table 2.1 Project Schedule

The table below presents the planned development schedule for the Artisa project. The project will follow an **Agile development methodology**, where the system will be developed incrementally through different phases, where each phase builds upon the previous phase. The initial phases focus on requirement analysis and system design, followed by frontend and backend development. The final phases are dedicated to testing, debugging, documentation, and project presentation to ensure the developed system meets the identified objectives.

Phase	Activities	Estimated Duration
Phase 1: Requirement Analysis	Identify system requirements, analyze existing platforms, define features, and conduct feasibility study.	Week 1–2
Phase 2: System Analysis and Design	Prepare UML diagrams, DFD, ER diagram, database design, and UI prototypes using Figma.	Week 3–4
Phase 3: Frontend Development	Develop user interfaces including homepage, marketplace, profiles, cart, and dashboards using React.js and Tailwind CSS.	Week 5–8
Phase 4: Backend Development	Implement Laravel backend, authentication, user roles, artwork management, orders, commissions, and APIs.	Week 7–11
Phase 5: Database Implementation	Create MySQL database, relationships, migrations, and integrate database with backend.	Week 8–11
Phase 6: Payment and Additional Features	Implement payment gateway, file storage, reviews, and additional system features.	Week 12–13
Phase 7: Testing and Debugging	Perform functional testing, fix errors, improve security, and optimize user experience.	Week 14
Phase 8: Documentation and Presentation	Prepare final report, screenshots, presentation slides, and project demonstration.	Week 15–16

REFERENCES

- Cloudinary Image & Video API Platform - Docs Home Page | Documentation.* (n.d.). Retrieved July 23, 2026, from <https://cloudinary.com/documentation>
- Etsy Singapore - Shop for handmade, vintage, custom, and unique gifts for everyone.* (n.d.). Retrieved July 23, 2026, from <https://www.etsy.com/>
- Fiverr | Freelance services marketplace | Find top global talent.* (n.d.). Retrieved July 23, 2026, from <https://www.fiverr.com/>
- Home - Hamrobazar.* (n.d.). Retrieved July 23, 2026, from <https://hamrobazaar.com/>
- Installation | Laravel 13.x - The clean stack for Artisans and agents.* (n.d.). Retrieved July 23, 2026, from <https://laravel.com/docs/13.x>
- Khalti Payment Gateway.* (n.d.). Retrieved July 23, 2026, from <https://docs.khalti.com/>
- Laravel Sanctum | Laravel 13.x - The clean stack for Artisans and agents.* (n.d.). Retrieved July 23, 2026, from <https://laravel.com/docs/13.x/sanctum>
- MySQL :: MySQL Documentation.* (n.d.). Retrieved July 23, 2026, from <https://dev.mysql.com/doc/>
- Nepal Rastra Bank (NRB) - the official site of the Central Bank of Nepal.* (n.d.). Retrieved July 23, 2026, from <https://www.nrb.org.np/>
- Online Shopping in Nepal: Best Deals, Prices & Discounts - Daraz.com.np.* (n.d.). Retrieved July 23, 2026, from <https://www.daraz.com.np/#?>
- React.* (n.d.). Retrieved July 23, 2026, from <https://react.dev/>
- Showcase, Discover, and Hire Creatives :: Behance.* (n.d.). Retrieved July 23, 2026, from <https://www.behance.net/>